CHEMISTRY MASTER SYLLABUS

CLASS 12 • NCERT / CBSE

Chapter 1: Solutions

- Types of Solutions
- Concentration of Solutions
- Solubility
- Vapour Pressure
- Raoult's Law
- Ideal Solutions
- Non-Ideal Solutions
- Azeotropes
- Colligative Properties
- Relative Lowering of Vapour Pressure
- Elevation of Boiling Point
- Depression of Freezing Point
- Osmotic Pressure
- Abnormal Molar Mass
- Van't Hoff Factor

Chapter 2: Electrochemistry

- Electrochemical Cells
- Galvanic Cells
- Electrode Potential
- Standard Electrode Potential
- Nernst Equation
- Cell Potential
- Gibbs Energy and Cell Potential
- Conductance
- Conductivity
- Molar Conductivity
- Kohlrausch's Law
- Electrolysis
- Batteries
- Fuel Cells
- Corrosion

Chapter 3: Chemical Kinetics

- Rate of Reaction
- Rate Law
- Rate Constant
- Order of Reaction
- Molecularity
- Integrated Rate Equations
- Half-Life
- Temperature Dependence of Rate
- Arrhenius Equation
- Activation Energy
- Collision Theory
- Catalysis

Chapter 4: d- and f-Block Elements

- Transition Elements
- Electronic Configuration
- General Characteristics
- Oxidation States

- Atomic and Ionic Radii
- Magnetic Properties
- Coloured Ions
- Catalytic Properties
- Interstitial Compounds
- Alloy Formation
- Lanthanoids
- Lanthanoid Contraction
- Actinoids

Chapter 5: Coordination Compounds

- Coordination Compounds
- Coordination Entities
- Ligands
- Coordination Number
- Nomenclature
- Isomerism
- Werner's Theory
- Valence Bond Theory
- Crystal Field Theory
- Bonding in Coordination Compounds
- Stability of Coordination Compounds
- Applications of Coordination Compounds

Chapter 6: Haloalkanes and Haloarenes

- Classification
- Nomenclature
- Nature of Carbon-Halogen Bond
- Preparation
- Physical Properties
- Chemical Properties
- Nucleophilic Substitution Reactions
- Elimination Reactions
- Haloarenes
- Polyhalogen Compounds

Chapter 7: Alcohols, Phenols and Ethers

- Classification
- Nomenclature
- Alcohols
- Phenols
- Ethers
- Preparation
- Physical Properties
- Chemical Properties
- Acidity of Phenols
- Reactions of Alcohols
- Reactions of Phenols
- Reactions of Ethers

Chapter 8: Aldehydes, Ketones and Carboxylic Acids

- Classification
- Nomenclature
- Aldehydes
- Ketones
- Carboxylic Acids
- Preparation

- Physical Properties
- Chemical Properties
- Nucleophilic Addition Reactions
- Oxidation Reactions
- Reduction Reactions
- Decarboxylation

Chapter 9: Amines

- Classification
- Nomenclature
- Preparation
- Physical Properties
- Chemical Properties
- Basic Character
- Diazonium Salts
- Diazonium Reactions

Chapter 10: Biomolecules

- Biomolecules
- Carbohydrates
- Monosaccharides
- Disaccharides
- Polysaccharides
- Proteins
- Amino Acids
- Peptides
- Protein Structure
- Enzymes
- Vitamins
- Nucleic Acids
- DNA
- RNA